Deep Learning Based Classification of Breast Ultrasound Images

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Abstract

Breast cancer is one of the most prevalent cancers affecting women worldwide. Early detection plays a critical role in improving patient survival rates. In this project, we develop a deep learning based image classification system for breast ultrasound images using the BUSI dataset. The objective is to classify ultrasound images into three categories: Normal, Benign, and Malignant.

A baseline convolutional neural network model based on ResNet18 is first trained without addressing dataset imbalance. Subsequently, different techniques for handling class imbalance are explored, including oversampling using a weighted sampler, data augmentation, and focal loss. The performance of these approaches is evaluated using standard classification metrics such as accuracy, precision, recall, F1-score, and confusion matrices. The results allow comparison of different strategies for improving classification performance on imbalanced medical imaging datasets.

1 Introduction

Breast cancer detection through medical imaging is an important application of artificial intelligence in healthcare. Among various imaging modalities, ultrasound imaging is widely used for breast examination because it is safe, cost-effective, and non-invasive.

Recent advances in deep learning have significantly improved performance in medical image analysis tasks such as classification, segmentation, and detection. Convolutional neural networks (CNNs) in particular have demonstrated strong capabilities in learning complex patterns from medical images.

However, medical datasets often suffer from class imbalance, where some classes contain significantly more samples than others. This imbalance can bias machine learning models toward majority classes, resulting in poor performance on minority classes.

In this project, we analyze the BUSI breast ultrasound dataset and develop a deep learning based classification system. We investigate the impact of several class imbalance handling techniques and compare their effectiveness.

2 Dataset

The dataset used in this project is the Breast Ultrasound Images (BUSI) dataset. It contains ultrasound images categorized into three classes:

- Normal
- Benign
- Malignant

Each image represents a breast ultrasound scan and some images also include segmentation masks identifying lesion regions. For this classification task, only the ultrasound images are used, while the segmentation masks are excluded.

2.1 Class Distribution

The dataset contains an imbalanced distribution of samples across the three classes. The class distribution is shown in Table 1.

Class	Number of Images
Normal	133
Benign	437
Malignant	210

Table 1: Class distribution of the BUSI dataset

Figure 1 shows the class distribution visualization.

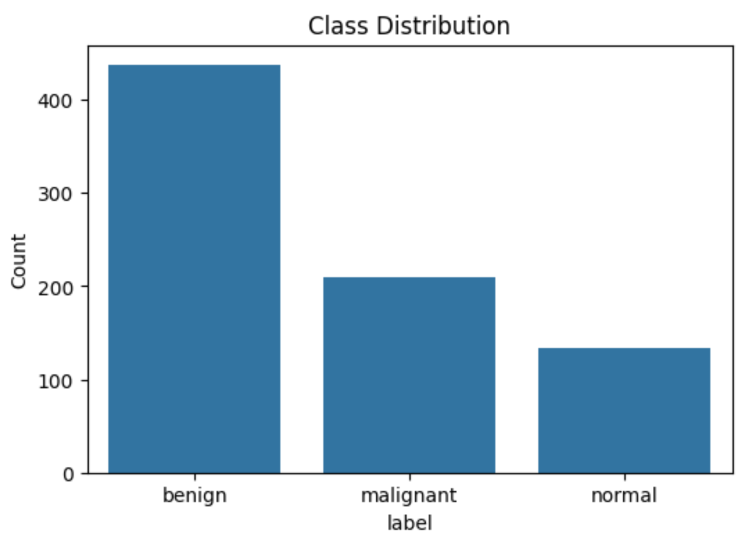


Figure 1: Class distribution of the BUSI dataset

The imbalance in the dataset motivates the exploration of techniques to mitigate its effect during model training.

3 Data Preprocessing

Several preprocessing steps were applied before training the deep learning models.

- All images were resized to 224×224 pixels to match the input size required by the pretrained CNN model.
- Images were converted into tensors and normalized.

- The dataset was split into training, validation, and test sets using stratified sampling to preserve class distribution.

The dataset split used in this project is shown below:

- Training set: 70%
- Validation set: 15%
- Test set: 15%

The training set is used to learn model parameters, the validation set is used for monitoring training performance and preventing overfitting, and the test set is used for final evaluation.

4 Methodology

The methodology followed in this project consists of several stages, starting from data preparation to model evaluation.

4.1 Dataset Preparation

The BUSI dataset was first organized by removing segmentation mask images and retaining only ultrasound images for classification. Each image was associated with one of the three labels: Normal, Benign, or Malignant.

The dataset was then split into training, validation, and test sets using stratified sampling in order to preserve the class distribution across all subsets.

4.2 Training Pipeline

The training pipeline consisted of the following steps:

- Image preprocessing and normalization
- Loading images using a PyTorch dataset and dataloader
- Training a convolutional neural network using transfer learning
- Monitoring performance on the validation set
- Evaluating final performance on the test set

4.3 Experimental Design

Four different experiments were conducted to analyze the effect of class imbalance handling techniques:

- Baseline model without imbalance handling
- Oversampling using a weighted sampler
- Data augmentation to increase training data diversity
- Training with focal loss to emphasize hard examples

All experiments used the same architecture and training pipeline in order to ensure a fair comparison between methods.

5 Model Architecture

In this project, we use the ResNet18 convolutional neural network architecture pretrained on the ImageNet dataset.

Transfer learning is applied by replacing the final fully connected layer of the network to produce predictions for three output classes corresponding to:

- Normal
- Benign
- Malignant

The model was trained using the Adam optimizer with a learning rate of 0.0001 and a batch size of 32. Cross entropy loss was used for the baseline model, while focal loss was used in the corresponding experiment designed to address class imbalance.

6 Imbalance Handling Techniques

To analyze the effect of class imbalance handling techniques, four different training strategies were explored.

6.1 Baseline Model

The baseline model is trained using the original dataset without applying any specific class imbalance handling technique. This provides a reference performance for comparison.

6.2 Oversampling

Oversampling was implemented using a WeightedRandomSampler during training. This technique increases the probability of selecting samples from minority classes, ensuring that the model is exposed to a more balanced distribution during training.

6.3 Data Augmentation

Data augmentation techniques were applied to increase the diversity of the training dataset and improve model generalization. The transformations used include:

- Horizontal flipping
- Random rotation
- Random affine transformations

These transformations simulate variations in ultrasound images and help the model learn more robust representations.

6.4 Focal Loss

Focal loss is designed to address class imbalance by reducing the contribution of easily classified examples and focusing training on harder samples.

The focal loss function can be expressed as:

$$FL(p_t) = -\alpha(1 - p_t)^\gamma \log(p_t)$$

where γ is the focusing parameter that controls how strongly the loss focuses on hard examples.

7 Evaluation Metrics

The performance of each model was evaluated using the following classification metrics:

- Accuracy
- Precision
- Recall
- F1 Score
- Confusion Matrix

These metrics provide a comprehensive evaluation of model performance, particularly in imbalanced classification tasks where accuracy alone may be misleading.

8 Results

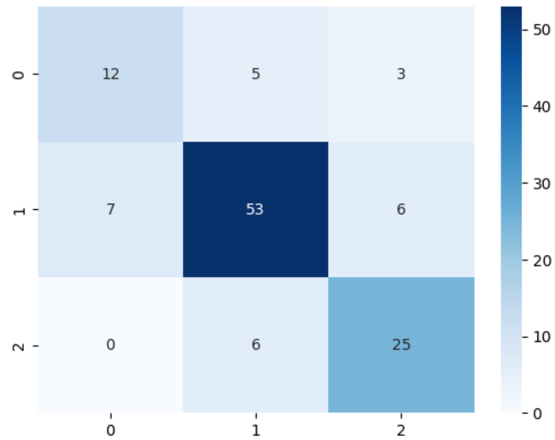
The test results obtained for different training strategies are summarized in Table 2.

Method	Accuracy	Precision	Recall	F1 Score
Baseline Model	0.77	0.73	0.74	0.73
Oversampling	0.84	0.85	0.77	0.79
Data Augmentation	0.81	0.81	0.76	0.78
Focal Loss	0.84	0.87	0.77	0.80

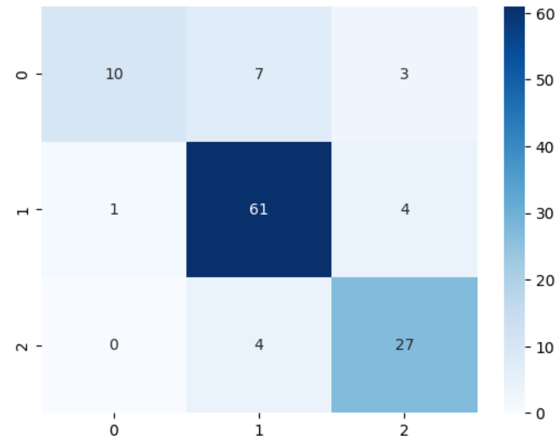
Table 2: Comparison of different imbalance handling techniques

8.1 Confusion Matrix

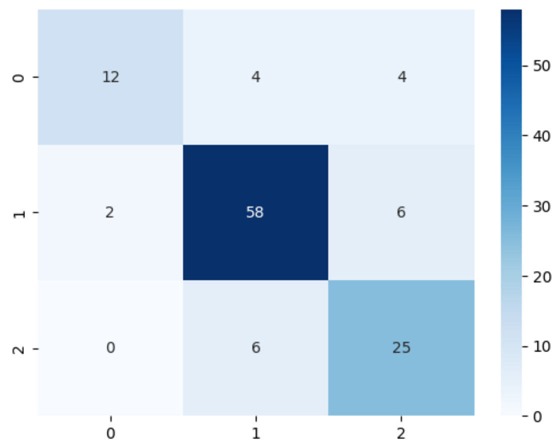
The confusion matrices for each model are shown below.



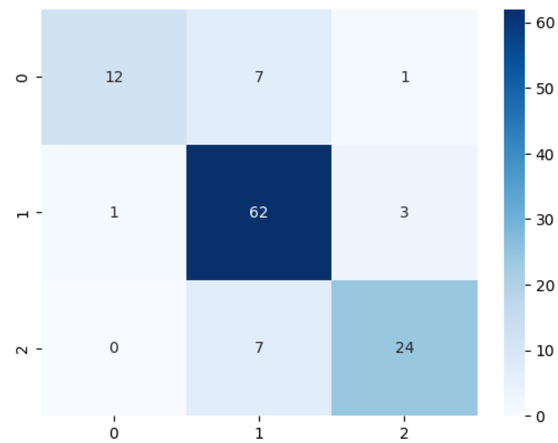
(a) Baseline Model



(b) Oversampling Model



(c) Data Augmentation Model



(d) Focal Loss Model

Figure 2: Comparison of Confusion Matrices for all models

9 Discussion

The experiments demonstrate that class imbalance significantly influences model performance. The baseline model achieves the lowest accuracy and F1 score, indicating that training on the imbalanced dataset without corrective techniques leads to reduced performance, particularly for minority classes.

Applying imbalance handling strategies improves the overall classification results. Oversampling increases the representation of minority classes during training and leads to a noticeable improvement in accuracy. Data augmentation also improves model generalization by introducing additional variations in the training data.

Among the evaluated techniques, focal loss achieves the best precision and F1 score, suggesting that focusing the learning process on difficult samples helps the model better distinguish challenging cases such as malignant tumors. Overall, the results indicate that incorporating imbalance handling techniques improves classification performance on the BUSI dataset.

10 Conclusion

In this project, we developed a deep learning based classification system for breast ultrasound images using the BUSI dataset. A ResNet18 model was trained as a baseline and compared with several imbalance handling techniques including oversampling, data augmentation, and focal loss.

Experimental results show that addressing class imbalance improves model performance compared to the baseline model. Among the evaluated methods, focal loss and oversampling achieved the best results, demonstrating the importance of handling imbalanced data in medical image classification tasks.